# `ascentopt`: a launch-trajectory / propulsion co-design tool

Modeling approach, verification, and a Starship Block 3 example

MECE 4305 Mechanics & Thermodynamics of Propulsion — Rocket module, Fall 2026

D. E. Tew (Columbia University) with Claude

prototype v0.1, 9 November 2026

**Abstract**

`ascentopt` is a multi-phase direct-collocation launch-trajectory optimizer built on Dymos/OpenMDAO. It carries the 3-DOF point-mass physics of a conventional ascent simulation, but thrust direction and throttle are solved for rather than prescribed, staging is a phase linkage, orbit insertion is a constraint and payload is the objective. Vehicle parameters (stage propellant loads, dry masses, reserves, engines) are an input deck that can be swept now and optimized in the same problem in the next milestone. This report documents the model, the numerical method, the verification performed, and a worked example: SpaceX Starship/Super Heavy Block 3 to a 275 km × 26° orbit from Starbase. With published propellant loads, Raptor 3 performance and estimated dry masses of 200/100 t, the optimizer places **293.3 t** in orbit on an expendable trajectory with losses of 1.83 km/s; holding back 8 % of the booster propellant for boost-back and 30 t in the ship for landing reduces this to 204.5 t, and the ship's dry mass trades against payload exactly 1:1. The results are grid-independent to 0.04 %, re-integrate explicitly to within 211 m and 4.5 m/s, and close the $\Delta v$-budget identity to 1 mm/s.

## 1 Why a trajectory optimizer

The standard preliminary treatment of a launch has two parts: a rule of thumb, $\Delta v_{\mathrm{LEO}} \approx v_{\mathrm{circ}} + \Delta v_{\mathrm{loss}} - v_{\mathrm{rot}} \approx 9.3\,\mathrm{km/s}$ with $\Delta v_{\mathrm{loss}} = 1.5$–$2.0\,\mathrm{km/s}$, and a forward simulation of a gravity turn that flies a prescribed steering law (vertical rise, pitch kick, thrust along the velocity) and reports where the vehicle ends up. Both are correct as far as they go, and neither can answer the questions a vehicle or mission design asks:

- How much payload does a *given* two-stage vehicle deliver to a *given* orbit, and where should it stage?
- How large are the gravity, drag, steering and back-pressure losses *for this vehicle*, rather than as a band?
- How much does each kilogram of booster dry mass, ship dry mass or return propellant cost in payload?
- What thrust, $I_{sp}$, chamber pressure and burn time must the propulsion system deliver — the inputs to thrust-chamber and cycle design?

Table 1 shows the size of the problem with the hand method. The rocket equation with the Starship deck of Section 5 returns anywhere from 276 to 339 t of payload depending on where in the 1.5–2.0 km/s loss band one lands; with the *actual* $\Delta v$ requirement from the optimized trajectory it returns 293.3 t, identically the optimizer's answer. The trajectory tool does not change the physics — it removes the assumption.

## 2 What the tool is

Figure 1 shows the workflow. A YAML vehicle deck and a mission definition are turned into a Dymos `Trajectory` (one phase per burn, linked in time and state); a prescribed gravity-turn law

**Table 1:** Rocket-equation closure for the Starship Block 3 deck (vacuum $I_{sp}$ 350.0/363.9 s, dry 200/100 t) versus the assumed $\Delta v$ requirement. The optimizer's requirement for this vehicle and orbit is 9,155 m/s.

| Assumed $\Delta v_{\text{req}}$ [m/s] | Loss assumption | Payload, no reserves [t] | Payload, 292/30 t reserves [t] |
|---|---|---|---|
| 8,800 | 1.5 km/s | 339.0 | 250.6 |
| 9,300 | 2.0 km/s (+ margin) | 276.2 | 195.7 |
| **9,155** | **optimizer: 1,833 m/s** | **293.3** | 210.7[†] |
| `ascentopt` (trajectory optimized) | | **293.3** | **204.5** |

[†] the reserve case has its own optimal trajectory with 1,876 m/s of losses; the hand method cannot know that.

provides the initial guess; IPOPT solves the resulting sparse nonlinear program; the optimized controls are re-integrated with an explicit integrator as a check; and the outputs are the trajectory time histories plus staging state, loss breakdown, and the propulsion requirements for the next design steps. Table 2 compares the capabilities with a prescribed-steering forward simulation — the reference implementation (`LaunchTrajectoryAnalysis.ipynb`) from which the tool was developed.

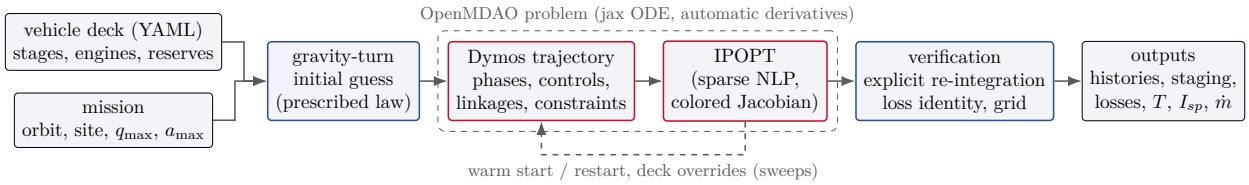



**Figure 1:** Workflow of `ascentopt`. A prescribed-law forward integrator and the downstream propulsion design steps bracket the optimizer: the former supplies the guess and an independent check, the latter consume the outputs.

# 3 Physical model

## 3.1 Equations of motion

The vehicle is a point mass in an Earth-centered inertial frame (Figure 2). With position $\boldsymbol{r}$, inertial velocity $\boldsymbol{v}$, propellant mass of the active stage $m_p$, and total mass $m = m_p + m_{\text{struct}} + m_{\text{pay}}$ (structure of the active stage plus everything above it),

$$\dot{\boldsymbol{r}} = \boldsymbol{v}, \tag{1}$$

$$\dot{\boldsymbol{v}} = -\frac{\mu}{|\boldsymbol{r}|^3}\boldsymbol{r} + \frac{T}{m}\,\hat{\boldsymbol{u}} - \frac{D}{m}\frac{\boldsymbol{v}_{\text{rel}}}{|\boldsymbol{v}_{\text{rel}}|}, \qquad \boldsymbol{v}_{\text{rel}} = \boldsymbol{v} - \boldsymbol{\omega} \times \boldsymbol{r}, \tag{2}$$

$$\dot{m}_p = -\tau\,\dot{m}_{\max}, \qquad D = \tfrac{1}{2}\rho(h)\,|\boldsymbol{v}_{\text{rel}}|^2 C_D A_{\text{ref}}, \qquad T = \tau\,(T_{vac} - p_a(h)\,A_e)\,. \tag{3}$$

The atmosphere co-rotates with the Earth, so drag acts on $\boldsymbol{v}_{\text{rel}}$ rather than on the inertial velocity; the rotation credit of 418 m/s at Starbase appears through $\boldsymbol{v}_0 = \boldsymbol{\omega} \times \boldsymbol{r}_0$ rather than as a separately signed term. $C_D = 0.5$ (constant) and $A_{\text{ref}} = \pi(4.5\,\text{m})^2 = 63.6\,\text{m}^2$; a Mach-dependent $C_D$ table is a planned refinement.

## 3.2 Environment and engine models

The US Standard Atmosphere 1976 is implemented as a table (analytic layers to 86 km, published values above) and evaluated as a log-linear interpolant, which keeps $\rho(h)$ and $p_a(h)$ smooth and differentiable to orbit (Figure 3). It reproduces the `ambiance` ISA package to within 0.1 % below 81 km.

An engine is specified by one anchor pair ($T_{vac}$, $I_{sp,vac}$) and an exit area; mass flow follows from $\dot{m}_{\max} = T_{vac}/(I_{sp,vac}\,g_0)$ and the sea-level thrust and $I_{sp}$ from the back-pressure term (Figure 4).

**Table 2:** Capability comparison with a prescribed-steering forward simulation (the reference implementation, `LaunchTrajectoryAnalysis.ipynb`).

| | Forward simulation | `ascentopt` v0.1 |
|---|---|---|
| Dynamics | 3-DOF point mass, spherical coordinates | 3-DOF point mass, ECI Cartesian (no polar singularity) |
| Earth rotation | initial velocity $\omega R\cos\phi$ | $\boldsymbol{v}_0 = \boldsymbol{\omega} \times \boldsymbol{r}_0$; drag on air-relative velocity |
| Atmosphere | `ambiance` ISA to 81 km | US76 to 1000 km, smooth, differentiable |
| Engines | $I_{sp}$ linear in $p_a/p_{SL}$ | $T(h) = \tau(T_{vac} - p_a A_e)$ from one anchor + exit area |
| Steering | prescribed gravity turn | solved for: unit vector $\boldsymbol{u}(t)$ and throttle $\tau(t)$ |
| Stages | one | any number, phase per stage, propellant state per stage |
| Constraints | none | $q \le q_{\max}$, $T/m \le a_{\max}$, $h \ge 0$, throttle bounds, orbit insertion |
| Objective | none (simulation) | max payload / min propellant / min time |
| Loss budget | — | five accumulator states; $\Delta v$-budget identity closed numerically |
| Verification | — | explicit re-integration, grid independence, benchmark (Longuski SSTO) |
| Run time | seconds | 2–4 min per case (IPOPT, 16 Radau segments per phase, laptop-class CPU) |

This matters because published Raptor figures are mutually inconsistent — Raptor 2's 230 tf SL / 327 s SL / 347 s vac triple *is* consistent with a 1.3 m exit ($p_a A_e = 134$ kN), whereas the often-quoted 258 tf vacuum thrust would require a 1.86 m exit — and a lookup table of mixed sources produces an unphysical engine. In the co-design milestone this block becomes an OpenMDAO group that produces $(T_{vac}, A_e, \dot{m})$ from chamber pressure, expansion ratio and mixture ratio through $c^*$ and $C_F$.

# 4 Optimal-control formulation and numerical method

## 4.1 The problem being solved

Table 3 lists the problem. Payload is a trajectory-level parameter shared by all phases and is the design variable the optimizer maximizes; the phase durations are free; the terminal set is a circular orbit of given altitude with free (or fixed) inclination.

**Table 3:** Optimal-control problem solved for the fixed-vehicle mode.

| | |
|---|---|
| Maximize | $m_{\text{pay}}$ (alternatives: minimize propellant used, minimize time) |
| over | $\hat{\boldsymbol{u}}(t)$, $\tau(t)$ in each burn phase; phase durations; $m_{\text{pay}}$ |
| subject to | dynamics of Section 3 in every phase (collocation defects) |
| | $\lvert\boldsymbol{u}\rvert^2 = 1$, $\tau_{\min} \le \tau \le 1$, $h \ge 0$, $q \le q_{\max}$, $T/m \le a_{\max}$ (path) |
| | continuity of $t$, $\boldsymbol{r}$, $\boldsymbol{v}$ and the loss states between phases (linkage) |
| | $m_p$ fixed to the stage load when a stage ignites; $m_p(t_f) \ge m_{\text{reserve}}$ at burnout (boundary) |
| | $h(t_f) = h_{\text{target}}$, $\gamma(t_f) = 0$, $\lvert\boldsymbol{v}(t_f)\rvert = \sqrt{\mu/(R_E + h_{\text{target}})}$ [, $i(t_f) = i_{\text{target}}$] |
| Initial state | $\boldsymbol{r}_0$ at the launch site, $\boldsymbol{v}_0 = \boldsymbol{\omega} \times \boldsymbol{r}_0$, loss states zero; first 10 s vertical at full thrust |

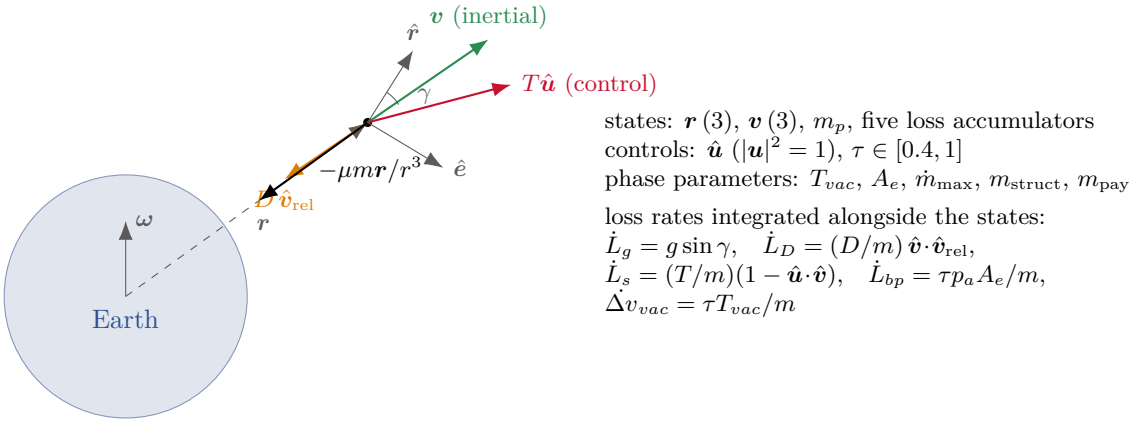



**Figure 2:** Forces on the point mass and the quantities carried as states. The loss rates are the terms of $d|\boldsymbol{v}|/dt = (T/m)\cos\alpha_T - D/m - g\sin\gamma$, so that $\Delta v_{vac} = (|\boldsymbol{v}_f| - |\boldsymbol{v}_0|) + L_g + L_D + L_s + L_{bp}$ holds identically — the $\Delta v$-budget identity closed numerically.

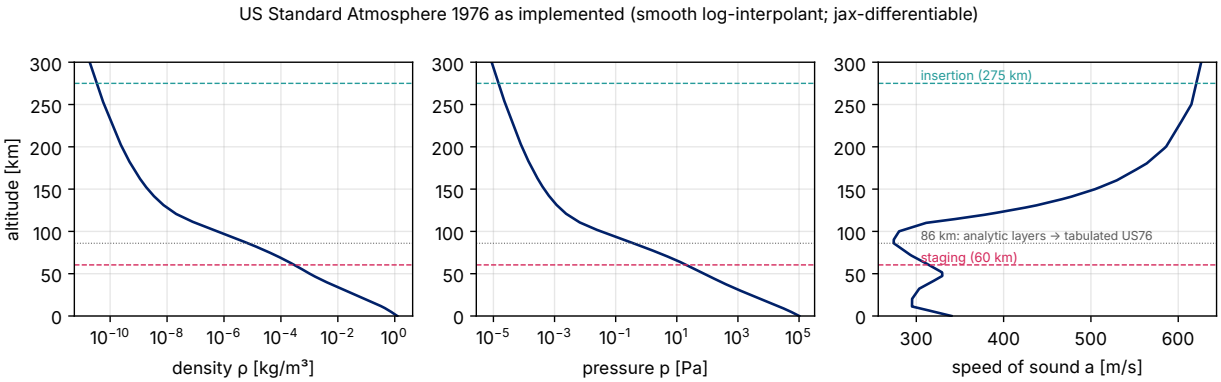



**Figure 3:** US76 atmosphere as implemented. Dashed lines mark the optimized staging altitude and the target orbit.

## 4.2 Phases and staging

Each stage burn is a Dymos phase with its *own* propellant state; the total mass is assembled inside the ODE from phase parameters (Figure 5). Position, velocity, time and the loss accumulators are linked between phases; propellant is not, so the mass drop at staging needs no linkage offset and the stage dry masses can later become design variables without touching the linkages. Return-propellant reserves enter as lower bounds on the propellant remaining at burnout.

## 4.3 Direct collocation

Each phase is discretized with the Legendre–Gauss–Radau pseudospectral transcription (Figure 6): the states are polynomials of degree 3 on each of 16 segments, the controls are polynomials on the same segments, and the dynamics are enforced as *defect constraints* at the collocation nodes. The result is a large, sparse nonlinear program (1,288 variables and 1,474 constraints for the two-stage case at 12 segments) rather than a sequence of simulations: the optimizer sees all of the trajectory at once, infeasible intermediate iterates are allowed, and the Jacobian is exact. Three implementation facts decided whether this converged:

1. **Derivatives.** The ODE is an OpenMDAO `JaxExplicitComponent`: partials come from automatic differentiation (checked against finite differences to $10^{-10}$). The node-diagonal sparsity of the partials must be *declared*, because OpenMDAO builds the total-Jacobian sparsity from declared patterns; undeclared, the coloring needed 504 colors for 843 variables, declared it needs 43.

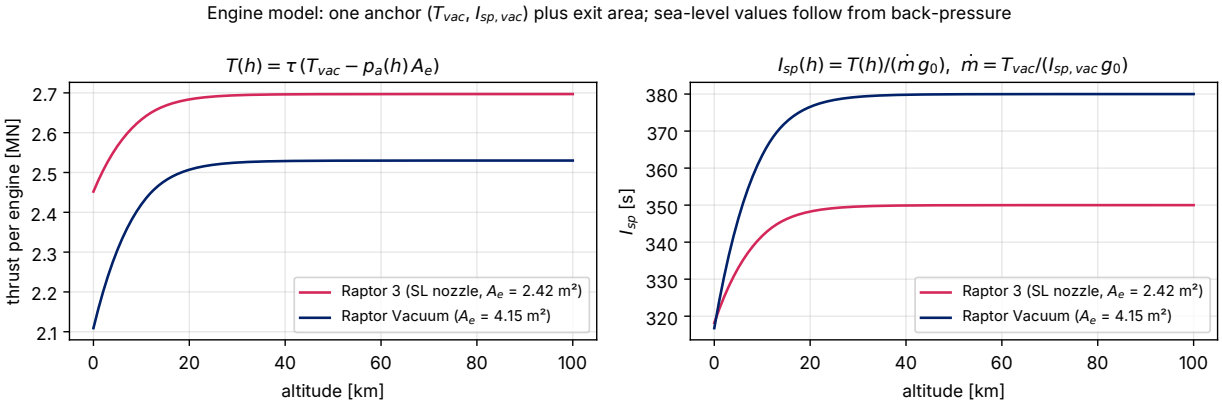



**Figure 4:** Engine model for the two Raptor variants of the Starship deck. Sea-level thrust and $I_{sp}$ are outputs, not inputs.

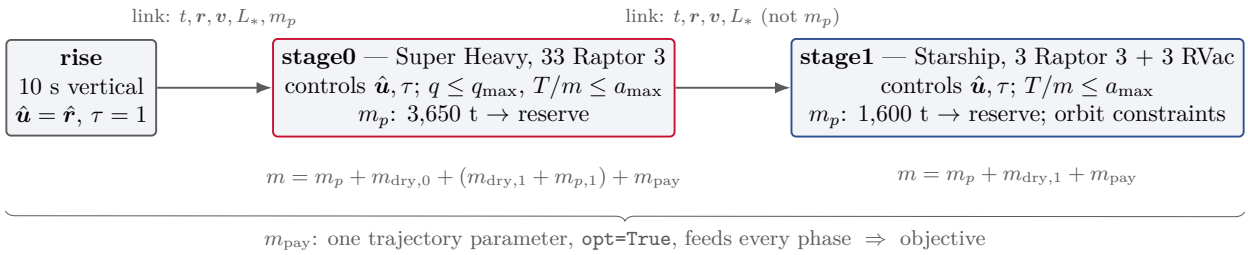



**Figure 5:** Phase structure. Hot staging is instantaneous in 3-DOF; the booster's reserve and the ship's landing propellant are inputs (not flown).

2. **Smooth outputs.** Flight-path angle, angle of attack and inclination are computed with $\arctan 2$; the $\arcsin$ / $\arccos$ forms have infinite slope exactly where a gravity-turn guess sits (thrust along velocity) and produced a dual infeasibility of $10^9$. The insertion constraint is $\gamma = 0$ rather than $\boldsymbol{r} \cdot \boldsymbol{v} = 0$, whose natural scale ($10^{10}\,\mathrm{m}^2/\mathrm{s}$) defeats the scaling.
3. **Optimizer.** IPOPT (through pyoptsparse/cyipopt, limited-memory Hessian, MUMPS). SciPy's SLSQP fails even Dymos's own single-stage benchmark; IPOPT solves it in 27 iterations.

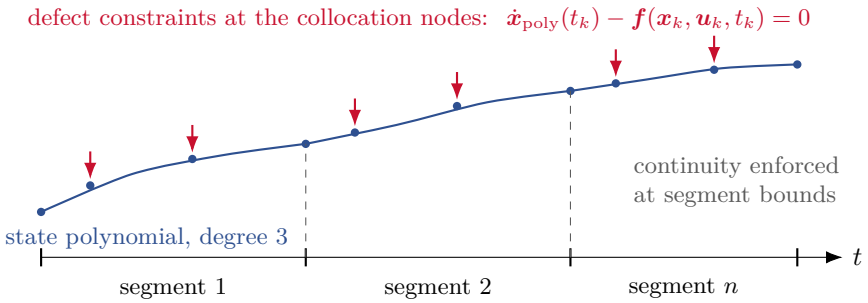



**Figure 6:** Direct collocation (Radau, order 3). States and controls are discretized on segments; dynamics become algebraic constraints solved simultaneously with the optimization.

## 4.4 Initial guess

A prescribed gravity-turn guidance law (vertical rise, pitch kick, thrust along the air-relative velocity) integrated with `solve_ivp` supplies the initial guess; the kick angle is tuned so that the flight-path angle is zero at burnout. For Starship this gives a trajectory that reaches $\gamma = 0$ at 218 km with 10.1 km/s — an eccentric orbit with 2.4 km/s of excess speed — which the optimizer

converts into payload. Cases are warm-started from the nearest solved case; a cold start of the baseline converges in about 260 IPOPT iterations and a warm-started sweep case in 20–50.

# 5 Example: Starship/Super Heavy Block 3 to 275 km × 26°

## 5.1 Inputs

Table 4 is the deck. Propellant loads and engine numbers are SpaceX's published Block 3 figures; the dry masses are *not* published; estimate ranges of 160–200 t (booster) and 85–100 t (ship) are used, their mid-points as the baseline, and swept. The mission is a "100 t to LEO" delivery from Starbase (26.0°N), 275 km circular (the altitude of Flight 14, the first orbital Starship flight, 28 Sep 2026), free inclination (eastward launch gives $i = 26.0°$), with $q \le 35$ kPa and $T/m \le 4\,g$.

**Table 4:** Vehicle and mission inputs (`vehicles/starship_v3_baseline.yaml`).

| Quantity | Super Heavy | Starship | Source / note |
|---|---|---|---|
| Propellant load | 3,650 t | 1,600 t | SpaceX Block 3 |
| Dry mass (baseline; swept) | 200 t | 100 t | not published; mid-points of 160–200 / 85–100 t estimates |
| Engines | 33 × Raptor 3 | 3 Raptor 3 + 3 RVac | SpaceX |
| Raptor 3 (SL nozzle) | 250 tf SL / 275 tf vac, $I_{sp,vac}$ 350 s ⇒ $A_e = 2.42$ m$^2$, $I_{sp,SL} = 318$ s | | SpaceX May 2026 / Aug 2024 |
| Raptor Vacuum | 258 tf, $I_{sp,vac}$ 380 s, 2.3 m exit diameter | | SpaceX (thrust, $I_{sp}$); exit diameter assumed‡ |
| Derived stage $T_{vac}$, $\dot{m}$ | 89.0 MN, 25,930 kg/s | 15.7 MN, 4,394 kg/s | engine model |
| Liftoff $T/W$ (sea level) | 1.46 | 0.89 at ignition (vac) | engine model |
| Throttle range | 0.40–1.00 | 0.40–1.00 | Raptor family |
| Diameter, $C_D$ | 9 m, 0.5 (constant) | | SpaceX; $C_D$ assumed |
| Reserves (baseline) | 0 | 0 | swept: 292 / 548 t booster, 30 t ship |
| Mission | Starbase 25.997°N, 275 km circular, inclination free, $q_{\max} = 35$ kPa, $a_{\max} = 4\ g$, 10 s vertical rise | | |

‡ a 2.4 m exit is also quoted for RVac; the RVac engines only burn above the 60 km staging altitude, where $p_a \le 22$ Pa and the thrust difference between the two exit areas is $\le 8$ N per engine (against 2.5 MN) — it does not affect the results.

## 5.2 Baseline solution

Figures 7–11 show the converged baseline (16 Radau segments per phase; IPOPT "solved to acceptable level", constraint violation $< 10^{-7}$). The optimizer finds **293.3 t** of payload. The booster throttles to 0.74 between 50 and 75 s to hold $q = 35$ kPa (the constraint is active — Figure 7, right-middle panel), returns to full thrust, and reaches the 4 $g$ acceleration limit exactly at MECO: $t = 144.7$ s, $h = 60.4$ km, $v = 2.61$ km/s inertial. The ship burns 463 s and throttles to its 40 % floor for the last 170 s; insertion is at $t = 607.7$ s with $h = 275.000$ km, $\gamma = 5 \times 10^{-10}$ deg, $|\boldsymbol{v}| = 7{,}740.26$ m/s, $i = 25.997°$. Both stages burn to their (zero) reserves. Table 5 summarizes the numbers and the $\Delta v$ budget.

## 5.3 What the payload depends on

Seven cases were run from the baseline by overriding deck values on the command line (Table 6, Figure 12). Three results stand out:

1. **Ship dry mass trades 1:1 with payload.** The four no-reserve cases are reproduced to 0.07 t by $m_{\text{pay}} = 425.8 - 0.161\, m_{\text{dry,booster}} - 1.001\, m_{\text{dry,ship}}$ [t] (Figure 13, left): a kilogram on the ship is a kilogram of payload, a kilogram on the booster costs 160 g.

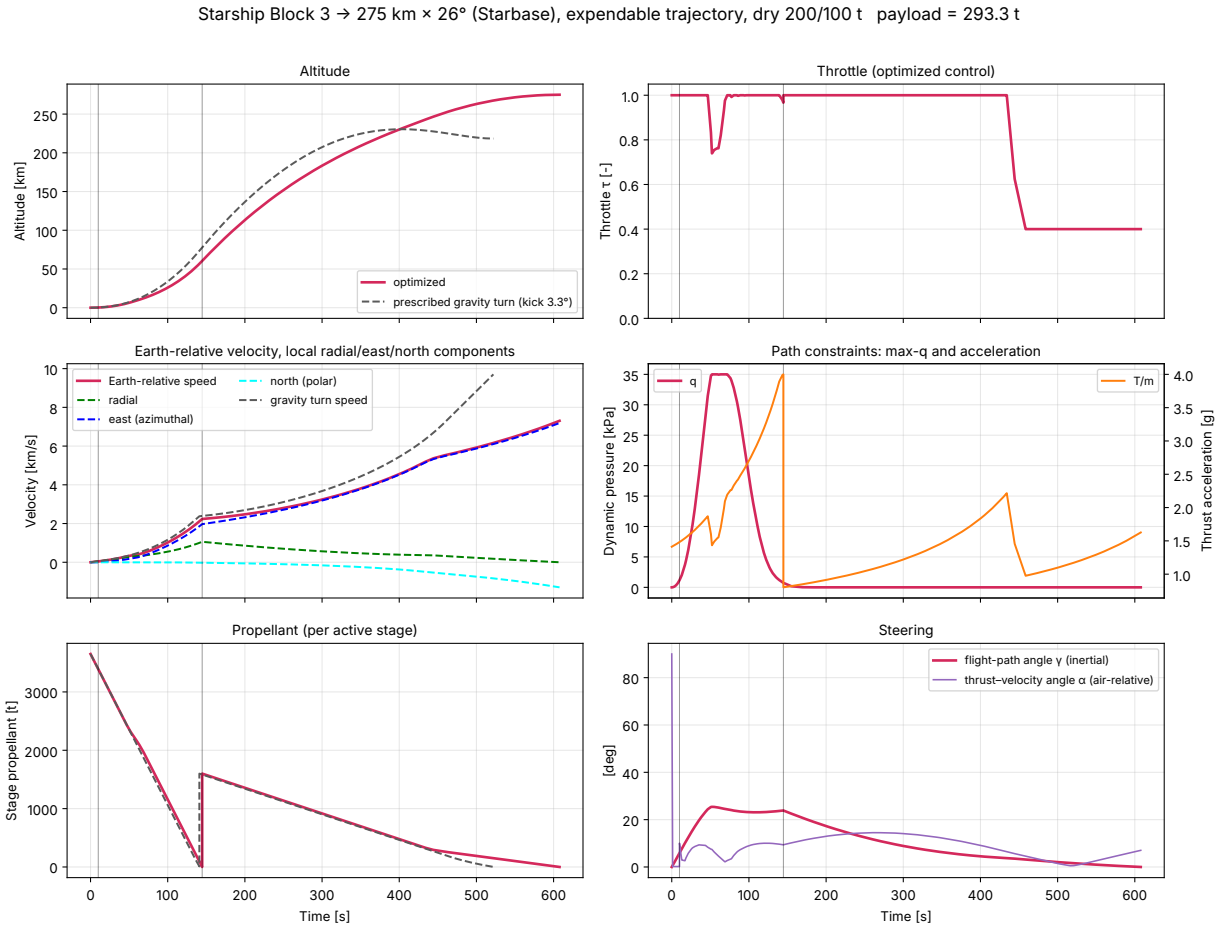



**Figure 7:** Baseline solution (red) against a prescribed gravity-turn law applied to the same vehicle (grey dashed). Left column: altitude, Earth-relative velocity components, stage propellant. Right column: throttle, the two active path constraints, flight-path angle and thrust–velocity angle. Vertical lines mark phase boundaries.

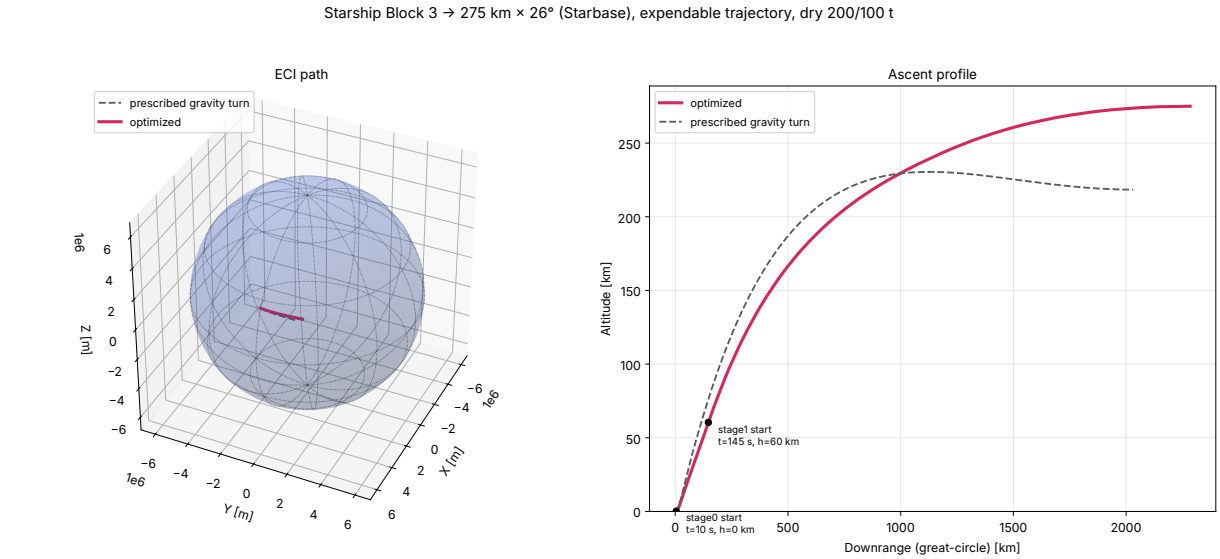



**Figure 8:** ECI path on the globe and the ascent profile. The gravity-turn law lofts to 218 km with 10.1 km/s (eccentric orbit); the optimizer inserts into the 275 km circle and carries the difference as payload.

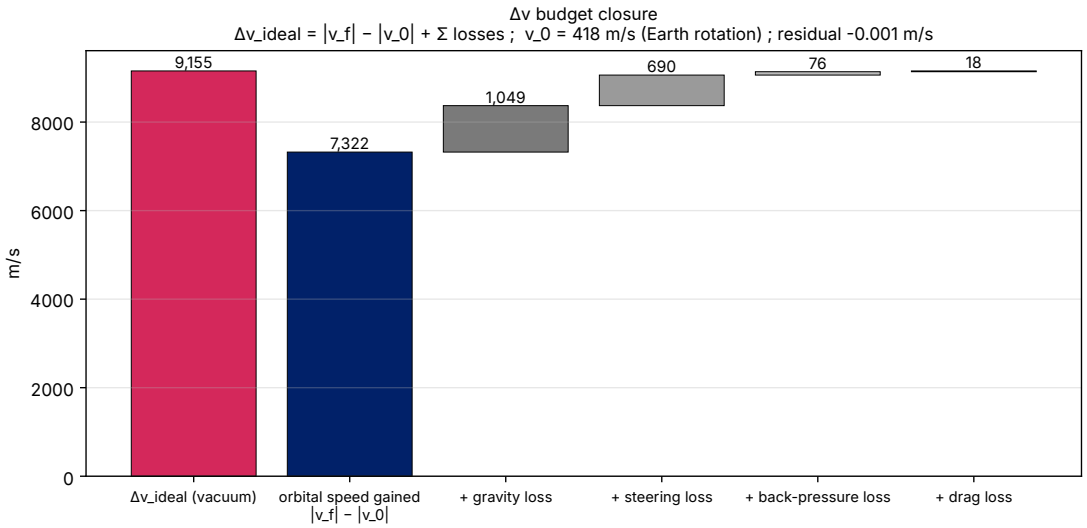



**Figure 9:** $\Delta v$ budget closure at the optimum. $\Delta v_{vac} = 9{,}155 = 7{,}322 + 1{,}049 + 690 + 76 + 18\,\mathrm{m/s}$; the residual of the identity is $-0.001\,\mathrm{m/s}$. Total losses 1,833 m/s, inside the textbook 1.5–2.0 km/s band.

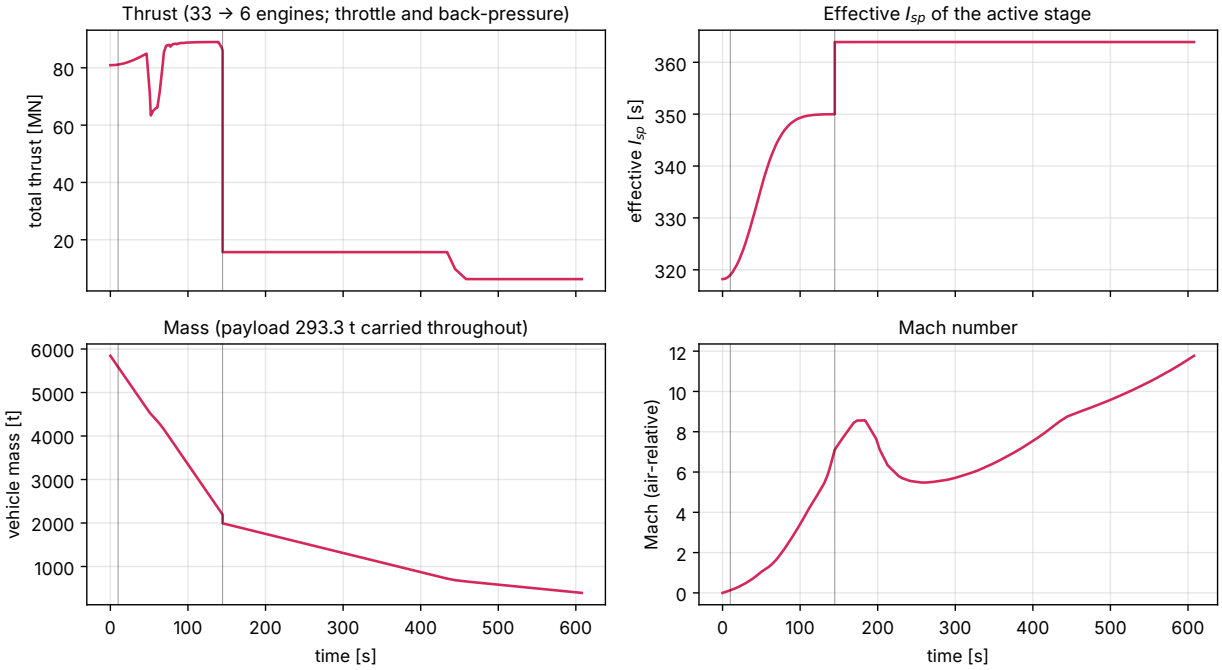



**Figure 10:** Thrust, effective $I_{sp}$ of the active stage, vehicle mass and air-relative Mach number. The $I_{sp}$ rise in the first 60 s is the back-pressure term; the step at 145 s is staging onto the Raptor 3/RVac set (thrust-weighted 363.9 s). Mach is shown for completeness only above ∼90 km.

**Table 5:** Baseline results (dry 200/100 t, no reserves).

| | |
|---|---|
| *Trajectory* | |
| Payload to 275 km × 26.0° | **293.3 t** |
| GLOW | 5,843 t |
| Staging (MECO) | 144.7 s, 60.4 km, 2.61 km/s inertial |
| Max-$q$ (constraint active) | 35.0 kPa at 50–75 s, booster throttle down to 0.74 |
| Max $T/m$ (constraint active) | 4.00 $g$ at MECO |
| Ship burn | 463 s; $\tau = 0.40$ for the last 170 s |
| Insertion | 607.7 s; $h = 275.000$ km, $\gamma = 5 \times 10^{-10}$ deg, $\lvert\boldsymbol{v}\rvert = 7{,}740.26$ m/s, $i = 25.997°$ |
| $\Delta v$ *budget* (m/s) | |
| $\Delta v_{vac}$ (both stages, vacuum-referenced) | 9,155 |
| orbital speed gained $\lvert\boldsymbol{v}_f\rvert - \lvert\boldsymbol{v}_0\rvert$ ($\lvert\boldsymbol{v}_0\rvert = 418$, Earth rotation at 26°N) | 7,322 |
| gravity loss $\int g \sin\gamma \, dt$ | 1,049 |
| steering loss $\int (T/m)(1 - \cos\alpha_T) \, dt$ | 690 |
| back-pressure loss $\int \tau p_a A_e / m \, dt$ | 76 |
| drag loss $\int D/m \, dt$ | 18 |
| identity residual | −0.001 |

2. **Return-propellant reserves are the largest lever.** Holding back 8 % of the booster propellant (292 t) plus 30 t in the ship costs 89 t of payload; 15 % costs 131 t. The optimizer moves the staging point by itself — from 145 s/60 km/2.61 km/s to 133 s/53 km/2.29 km/s at 8 % and 123 s/47 km/2.00 km/s at 15 % (Figure 14) — which is the mechanism behind the "stage low for reusability" rule.
3. **The gap to the published 100 t is attributable.** Even the pessimistic corner (dry 250/120 t plus 8 % reserve) gives 178 t. What remains is outside this model: the actual ship dry mass with heat shield and flaps (likely well above 120 t), boost-back propellant that grows with staging speed and must deliver a specific return trajectory rather than an 8 % allowance, engine performance margins and operational margins. The tool's contribution is to make the gap a list of named quantities with known sensitivities.

**Table 6:** Sweep results. All cases converged (IPOPT acceptable level, constraint violation $\leq 10^{-7}$); iterations are for the warm-started final chunk.

| Case | Payload [t] | Stage $t$ [s] | $h$ [km] | $v$ [km/s] | $L_g$ [m/s] | $L_s$ [m/s] |
|---|---|---|---|---|---|---|
| dry 160/85 t (low estimate), no reserves | 314.9 | 144.6 | 61.1 | 2.65 | 1,045 | 680 |
| **baseline: dry 200/100 t, no reserves** | **293.3** | 144.7 | 60.4 | 2.61 | 1,049 | 690 |
| dry 250/120 t, no reserves | 265.3 | 144.4 | 59.8 | 2.57 | 1,056 | 702 |
| ship dry 150 t (booster 200 t), no reserves | 243.3 | 144.7 | 60.4 | 2.61 | 1,049 | 690 |
| reserves: booster 292 t (8 %), ship 30 t | 204.5 | 133.4 | 53.0 | 2.29 | 1,083 | 716 |
| dry 250/120 t + reserves 292/30 t | 178.4 | 133.1 | 52.4 | 2.25 | 1,090 | 727 |
| reserves: booster 548 t (15 %), ship 30 t | 162.7 | 123.3 | 46.5 | 2.00 | 1,126 | 753 |

# 6 Verification

Table 7 lists what was checked. The explicit re-integration (Figure 15) restarts from the collocation state at each phase boundary and drifts by 211 m and 4.5 m/s by insertion; the drift originates at the throttle transition at 440 s, where the piecewise-polynomial control is least accurate, and halves from 12 to 16 segments. Grid refinement at that transition is the obvious next step and

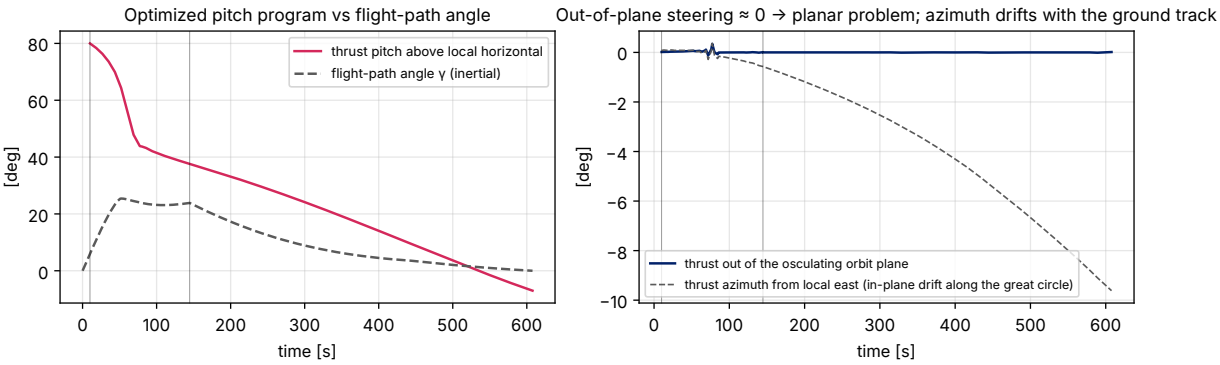



**Figure 11:** The optimized steering. Left: thrust pitch relative to the local horizontal and flight-path angle. Right: the component of thrust out of the osculating orbit plane is zero to plotting accuracy — the optimizer discovers that an eastward launch is a planar problem — while the azimuth measured from local east drifts as the ground track leaves the 26°N apex of the orbit.

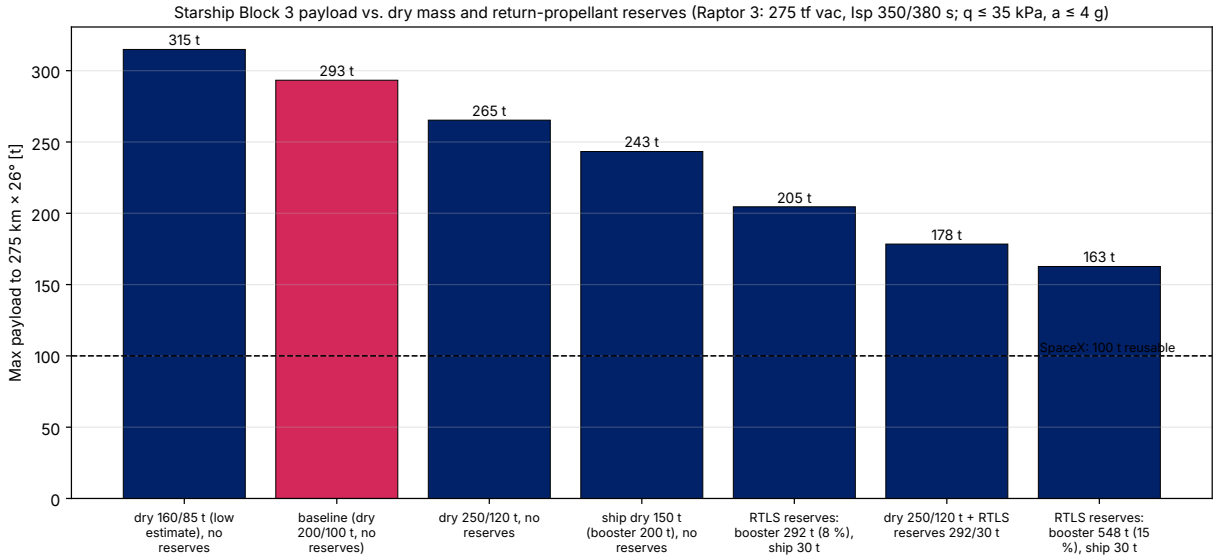



**Figure 12:** Maximum payload across the sweep. The dashed line is SpaceX's published reusable figure.

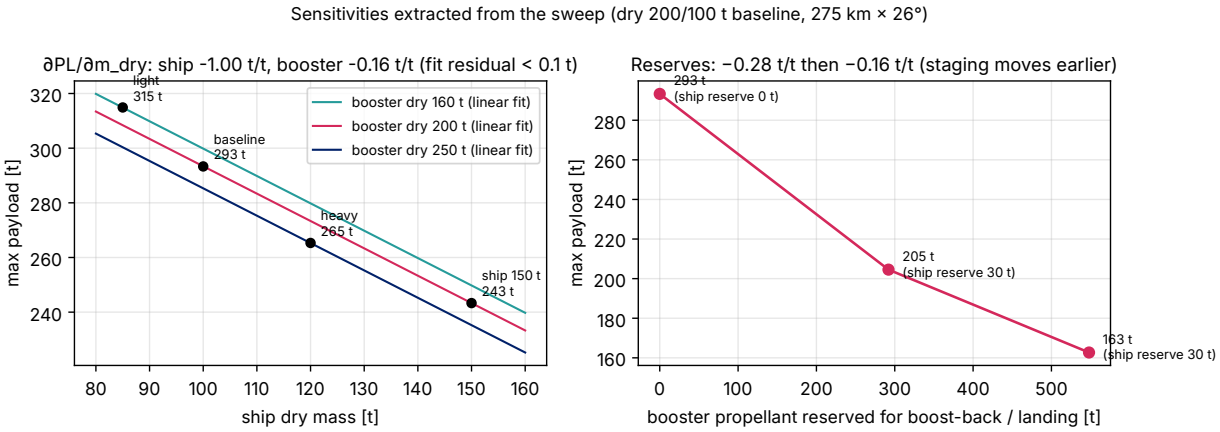



**Figure 13:** Sensitivities extracted from the sweep. Left: a linear model in the two dry masses reproduces the four no-reserve cases to 0.07 t. Right: payload against booster propellant held back for return; the slope flattens because the optimizer re-stages earlier.

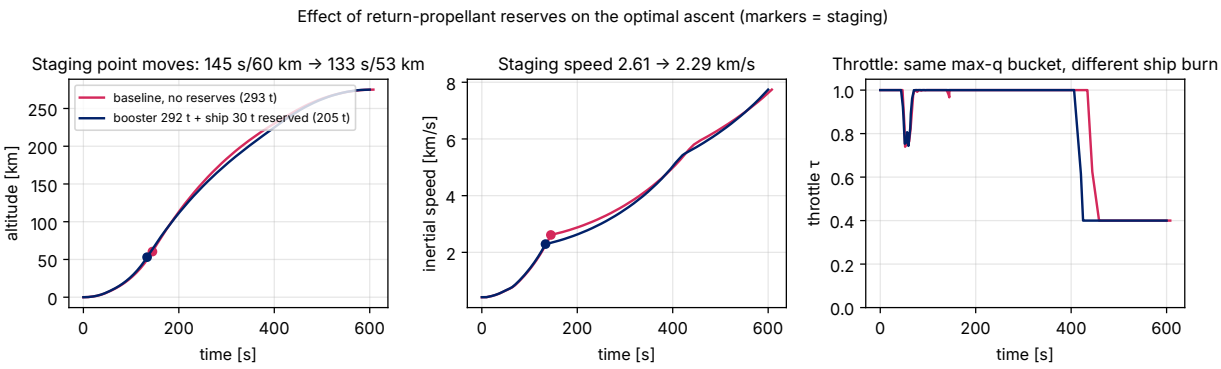



**Figure 14:** Baseline versus the 8 %-reserve case. The max-$q$ throttle bucket is unchanged; the staging point moves earlier and lower and the ship carries more of the $\Delta v$.

does not change the payload at the 0.04 % level.

**Table 7:** Verification performed on 9 Nov 2026.

| Check | Result |
|---|---|
| Benchmark: Dymos SSTO (Longuski Titan II min-time) with IPOPT | $t_f = 143.77$ s in 27 iterations (documented $\approx 143$ s); SciPy SLSQP failed in 3,000 iterations |
| Explicit re-integration of optimized controls (`simulate`) | 16 segments: +211 m, +4.5 m/s, 0.0002° at insertion; 12 segments: +322 m, +8.8 m/s |
| Grid independence | 293.20 t (12 segments) vs 293.31 t (16 segments) |
| Loss identity $\Delta v_{vac} = \Delta\lvert\boldsymbol{v}\rvert + \sum L$ | residual $\leq 0.004$ m/s in all seven cases |
| ODE partials (jax) vs finite differences | agree to $\sim 10^{-10}$ |
| US76 table vs the `ambiance` ISA package | $< 0.1$ % in $\rho$, $p$ below 81 km |
| Rocket-equation consistency | with the optimizer's $\Delta v_{vac}$ the hand calculation returns the optimizer's payload (Table 1) |
| Flight data (IFT-3 public timeline: staging 72 km, T+170 s, return-to-site profile) | 53 km/133 s with 8 % reserve, 47 km/123 s with 15 % — same regime; lower because reserves are allowances, not a flown return |

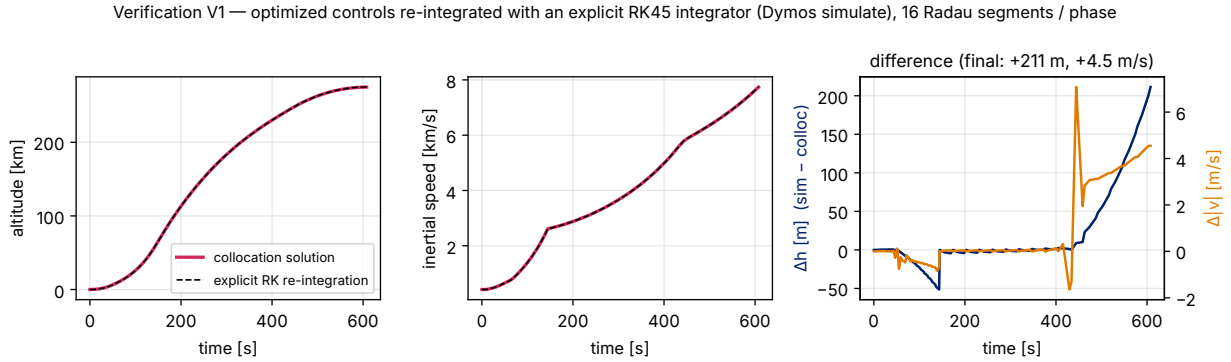



**Figure 15:** Collocation solution versus explicit RK45 re-integration of the optimized controls. Right: the difference, which starts at the 440 s throttle transition.

# 7 Limitations and next steps

- **Aerodynamics.** $C_D$ is constant; a Mach-dependent table is the first refinement (affects drag loss and the max-$q$ bucket, both small here).
- **Gravity, winds.** Point-mass spherical Earth, no $J_2$, no winds, no 6-DOF.
- **Reusability.** Reserves are inputs; boost-back and landing are not simulated, so the reserve–staging-speed coupling that drives real return-to-site designs is not closed.
- **Solution structure.** The ship's final 40 %-throttle arc is what IPOPT returned on this grid; it should be re-checked with grid refinement and a different initial guess before drawing conclusions from it.
- **Unphysical vehicles.** A "Falcon-9-like" single-stage test vehicle ($T/W = 3.75$, 95 % propellant fraction) did not converge as a max-payload problem; a feasibility run reached 250 km with $\approx 0$ payload and 79 $g$ at burnout. It is not a useful test case.
- **Co-design (milestone 2–3).** Engine model from $(p_c, \varepsilon, O/F)$ via $c^*$ and $C_F$, stage mass model, and the FFSC-cycle/turbopump models as OpenMDAO components in the same problem — the reason Dymos was chosen. In-space mission module (Hohmann/Lambert, $C_3$, aerocapture) for the Mars, Moon and Titan project missions.

# 8 Using the tool

```
conda env create -f environment.yml && conda activate ascentopt
python scripts/run_case.py --vehicle vehicles/starship_v3_baseline.yaml --out runs/
    starship \
    --h-target 275e3 --lat 25.997 --q-max 35e3 --a-max 39.2266 --segments 16 --max-iter
        400 --simulate
python scripts/run_case.py ... --out runs/heavy --warm-from runs/starship/*_out/
    dymos_solution.db \
    --set stage0.dry_mass=250e3 --set stage1.dry_mass=120e3
python scripts/make_figures.py runs/starship --out results/figs
bash reproduce_demo.sh # baseline + six-case sweep + figures
```

The environment is conda-forge (OpenMDAO 3.41, pyoptsparse 2.17, cyipopt 1.7/Ipopt 3.14, jax 0.7) plus dymos 1.16 from PyPI. `cyipopt` must be listed explicitly or the IPOPT driver fails to import. Results land in `result.json` next to the Dymos record `dymos_solution.db`; `-restart` resumes a case and `-warm-from` starts a new case from another case's record with the new deck re-applied.

## Sources

- SpaceX Starship and Raptor data as compiled on Wikipedia (Block 2/3 propellant loads, engine counts, Raptor 3 250/275 tf and 350 s, RVac 258 tf and 380 s, Flight 14 orbit) and in the New Space Economy review of Starship V3 (April 2026). The RVac exit diameter (2.3 m) and the dry-mass ranges are assumptions, as marked in Table 4.
- Dymos documentation: SSTO Earth-launch example (Longuski Titan II benchmark), Trajectory API; OpenMDAO `JaxExplicitComponent`.
- IFT-3 timeline (hot staging at T+2:50, 72 km): Cité de l'espace step-by-step report.